Round 13 | Carried fourth cumulants

One explicit mechanism, costed and falsified
Astra research note | 25 September 2026

Verdict: I do not have a mechanism that meets your gate. A full-depth dense pair core has a genuine all-distinct sector and a derived arithmetic cost of **15.981 u [D]**. I tested it against your stored public-network teacher: its deep standardized-kurtosis correlations are negative [Q-test]. It should be rejected, not presented as a solution.

Number labels: [D] = derived algebra/arithmetic; [Q] = quoted from your prompt or project evidence; [Q-test] = quoted from the new test log embedded in this PDF; [E] = estimate. Constants and dimensions in displayed equations are [D]; layer/source labels are indices. No estimated correlation is presented as a measurement.

The single candidate: a pair core carried from the first ReLU

Keep a symmetric dense matrix **G** and a dense propagator $\mathbf{U_l}$, each n by n [D]. Use the chain's existing first post-ReLU pair slices as the birth:

```
G_aa = k4(h_1)_aaaa
G_ab = 3 K22(h_1)_ab   (a != b)
U_2 = W_1;   U_(l+1) = U_l D_l W_l
D_l = diag(Phi(mu_l / sigma_l))
```

Later births are **zero by definition**. The slopes come from the baseline chain. Thus this is an explicit approximation with long memory, not an exact cumulant update through ReLU. Under your row-vector convention, U_l maps the first post-ReLU coordinates to pre-activation layer l.

```
F_l(v) = ((U_l v) * (U_l v)).T G ((U_l v) * (U_l v))
S_l = U_l * U_l;   V_l = G S_l
k4hat_l = colsum(S_l * V_l)
```

Here * means elementwise multiplication. At birth, the coefficient of $v_a^2 v_b^2$ is $2G_{ab} = 6K22_{ab}$ [D], which fixes the normalization. After dense transport, expanding F produces mixed monomials $v_a v_b v_c v_d$ with all indices distinct. G is unrestricted and dense; this is neither diagonal-only transport nor a covariance scale-mixture. It does discard other first-birth sectors and all subsequent nonlinear replenishment.

How it would enter the existing mean closure

It would **replace** the regenerated pre-activation diagonal in the existing Edgeworth term. Holding the other channels fixed, with alpha = mu/sigma:

```
delta mean_i = phi(alpha_i) (alpha_i^2 - 1)
               * (k4hat_i - k4regen_i) / (24 sigma_i^3)
```

This is the linear fourth-cumulant contribution [D]. The probe tested the auxiliary carrier only; it did not feed its failed prediction back into the means or covariance.

Arithmetic at the real shapes

All costs on this page are [D], unless marked otherwise. n = 1024 [Q]; one u = 2^{31} scalar FLOPs [Q].

The state uses $2n^2$ scalars: **8 MiB** in float32 [D], excluding readout buffers and multiplication workspace. G is fixed. One recurring transition plus readout uses exactly these shapes:

Operation	Shape	Scalar arithmetic
Scale U by slopes	1024 by 1024	n^2
Transport (UD)W	(1024 by 1024)(1024 by 1024)	One square product
Form S = U * U	1024 by 1024	n^2
Form V = GS	(1024 by 1024)(1024 by 1024)	One square product
Multiply S * V; reduce columns	1024 by 1024 to 1024	$2n^2$ - n

For an explicit, standard Strassen recursion of depth 5 with classical 32 by 32 leaves, let B be the scalar FLOPs of a square product. This uses 18 block additions per internal node; it does not assume the prompt's faster Winograd implementation.

```
B = 7^5 (2*32^3 - 32^2)
    + 18 sum_(j=0..4) 7^j (1024/2^(j+1))^2
  = 1,181,223,936 FLOPs = 0.550050259 u
```

Stage	Exact scalar FLOPs [D]	Cost [D]
Recurring transition + readout	2B + $4n^2$ - n = 2,366,641,152	1.102053165 u
First readout at layer 2 (U is copied from W)	B + $3n^2$ - n = 1,184,368,640	0.551514626 u
Whole network, including G = 3K22	29B + $60n^2$ - 15n = 34,318,393,344	15.980747223 u

There are 14 transported transitions and 15 readouts [D]. Fully classical products instead cost 29.015129566 u [D]. The diagnostic used classical float64 operations; it did not benchmark a metered Strassen implementation. Copies, allocations, peak workspace, and meter-specific charges are excluded. Phi and the birth slices are reused from the existing chain.

Budget ambiguity: this is two square-product calls per recurring layer [D], so it fails a literal cap of 1.5 calls [Q]. It fits the separate 20-25 u arithmetic allowance [Q]. Whether a deployed implementation fits the billed allowance remains unmeasured; arithmetic cost alone is not a billing guarantee.

Why long memory does not imply the requested correlation

The exact one-step carry oracle starts from the true, continually replenished tensor. This candidate starts from a restricted birth and thereafter uses only linearized transport. The oracle therefore cannot establish its accuracy. Adding arbitrary later births into this same G is not an exact closure; keeping separate propagated births changes this cost ledger.

Parity correction [D]: with an upstream tensor fixed, a fourth-order contraction contains four factors of the current W and is even under W to -W. The odd-parity argument for carried third cumulants cannot be applied unchanged to fourth cumulants. The prompt's empirical failure of cheap even-statistic predictors [Q] remains evidence, but is not such a parity theorem.

One falsifier, executed

Public MLP #0 | full history from the first birth | no teacher feedback

Run the equations above with stored baseline mu and variance at every layer. Compare k4hat with the teacher on $|mu_{true}/sigma_{true}| < 1.5$ [Q]. To match the project's pattern diagnostic, correlate excess kurtoses $k4hat/var_{true}^{2}$ and $k4_{true}/var_{true}^{2}$. Also report raw-diagonal correlation, so the metric convention cannot hide the failure. The required threshold is 0.7 [Q].

Layer	Pattern corr. candidate	Pattern corr. baseline	Raw k4 corr. candidate	Mean excess level ratio
12	-0.347	0.339	0.641	1.84%
13	-0.217	0.281	0.655	1.47%
14	-0.311	0.298	0.628	1.29%
15	-0.275	0.264	0.651	1.17%
16	-0.269	0.324	0.636	0.97%

Every table entry is [Q-test]. The level ratio is mean(predicted excess kurtosis) divided by mean(teacher excess kurtosis), on the same mask. It is not a norm ratio. There are 323-442 decisive neurons in these layers [Q-test].

Result: standardized pattern correlation is -0.347 to -0.217 [Q-test]; raw-diagonal correlation is 0.628-0.655 [Q-test]. Only 0.97-1.84% [Q-test] of the deep excess-kurtosis level survives in this candidate. It fails under either correlation convention. A positive rescaling cannot repair a negative standardized correlation [D].

The synthetic network was not run because failure on the public network already rejects a gate requiring both. This is a baseline-driven, full-history auxiliary test, not a teacher-forced one-step test and not a fully coupled end-to-end run. No final-mean MSE improvement is claimed.

Reproducibility

The PDF embeds **pair_carry_probe.py** and **pair_carry_probe.json**. Extract them with a PDF attachment viewer, then run `python pair_carry_probe.py`. The script uses NumPy and scipy.special.ndtr; its project-root path is explicit. It reads only these stored project files:

```
data/dumps/r7_cap_L1.npz
data/dumps/errbudget_v29plain.npz
data/kaggle_out/kaggle_out_k4pub/k4teach/weights_seed0.npz
data/kaggle_out/kaggle_out_k4pub/k4teach/layer{L}_seed0.npz
```

The first file supplies K4 and K22; the second supplies baseline L{l}_cmu, L{l}_cvar and L{l}_ck4; teacher files supply mean, C and K4d only for scoring. The script writes its JSON beside itself. Research/source files were not edited.

What this establishes

I reject this single mechanism. It demonstrates that a cheap, full-depth object with genuine all-distinct entries can still lose the needed memory because the birth rule and nonlinear evolution are inadequate. It does **not** prove that every inexpensive nonlinear state is impossible. I cannot justify a qualifying replacement from the available evidence, and I will not claim the requested correlation without a passing falsifier.

Sources: your attached Round 13 prompt [Q]; `consultations/fable/ROUND12_opus_report.md` [Q] for the carry-oracle context and pattern convention; embedded test log [Q-test]. Linear tensor transport background: ARC, mlp_cumulant_propagation. Equations and costs here are derived, not performance claims from that repository.